Keep it running and secure

IoT · Level 4 · Lesson 3



Today we will

- Plan maintenance: batteries, cleaning, recalibration, and a heartbeat that shows a silent failure.
- Secure devices: new passwords, updates, a separate network, only the data needed.
- Hand the system over so it outlives the project team.

Keep it alive

- Upkeep on a written rota
- A heartbeat and a last-seen time
- New passwords, updates, its own network
- Hand over before anyone leaves

Failure hunt

1. Warm-up (5 min): The garden sensor that said 40% all holiday.
2. Teach (8 min): Upkeep, the heartbeat, Mirai and the security habits.
3. Hunt (12 min): Groups list failures: how we'd notice, who fixes, prevention.
4. Share (7 min): Groups add their best three to a class table.
5. Guide (8 min): Write the device's one-page guide together.

Teach defence only: never try passwords on devices you don't own. Before class, find out who looks after the school's router, so the guide names a real person.

Let’s discuss

1. Which failure would your school notice last, and why?
2. Where should the school keep the project's passwords?
3. Who in the year below could join the team now?

Exit ticket

- Name one silent failure, and say how a heartbeat would reveal it.